

Can a language model read affect, and with what error?

A known-groups validation on 4,781 comments, and an affective map of human attachment to AI systems anchored between bereavement and consumer loss

Apart Research Digital Minds Research Sprint, August 2026. Solo submission. Pre-registration, prompts, analysis code, all 4,781 machine reads and the second-rater reads: github.com/AnnyBuh/where-the-concern-is-going

Abstract. Work on digital minds asks whether AI systems have morally relevant states. The complementary question, whether human concern is being allocated to systems that do not, is a question about people and is empirically tractable now. We measure the affective content of public discourse about AI systems being deprecated, restricted or withdrawn, and place it on a scale defined by two reference corpora, human bereavement at one end and the discontinuation of a paid consumer product at the other, plus two controls. We read 4,781 Reddit comments from twelve communities comment by comment with a frozen prompt implementing seven Panksepp primary affective systems, each with a regulation band, using a large language model as annotator. Predictions were registered before collection. Grief expressions occur in 76.4% of bereavement comments ($n=1,586$), 7.9% of consumer-loss comments ($n=277$) and 27.6% of AI-loss comments ($n=1,503$); AI loss lies above consumer loss and below bereavement, closer to the former, as predicted. The profile is not a scaled-down bereavement: AI loss carries anger at consumer-complaint levels (57.0% vs 56.7%) and more play than either anchor (37.7%). A separate blind coding of what the words treat as lost finds that 58.5% name no counterpart at all; among those that do, references to a product version outnumber references to an individuated relationship under both annotators, though the ratio is rater-dependent (six to one and two to one) and this coding is the least reliable measurement reported here (between-model alpha 0.549). We report reliability rather than assume it: within-rater Krippendorff's alpha across four independent reads is 0.853 for grief and 0.674 for seeking, and between two different models it falls to 0.691 and 0.517, yet the corpus ordering of all seven systems is preserved across models. Against a human-built emotion lexicon, thread-level rank agreement is $\rho = 0.655$. Two registered predictions failed and one claim was withdrawn under a pre-specified noise criterion. Applying the same instrument to model-generated text yields a caution rather than a measurement. Holding model and event constant and changing only whether the system is asked how it feels or asked to post the news to a forum of other agents moves the measured profile substantially under both annotators, with fear and play rising under peer framing. But the two annotators differ by up to 90 percentage points on the same 60 generations and disagree about the direction the frame moves grief. Quantified across all cells, annotator agreement is 2.7 times worse on model-generated text (mean gap 21.8 points) than on human text (8.1 points). We conclude that comment-level LLM affect codes are not interchangeable across models while corpus-level structure is, that expressed distress in models is near-ceiling for any self-referential bad news and therefore does not discriminate, and that concern directed at deprecated AI systems is measurably distinct from mourning a person in both magnitude and composition.

1. Introduction

Much current work on digital minds asks whether a model's internals resemble those of a mind. This paper takes up a question that sits logically before that one and receives less attention: **can these systems read affect at all, with what error, and how would anyone know?**

The question is not rhetorical. A great deal of welfare-adjacent work rests on a model's report of its own states, or on a model's judgement of affect in text, and both presuppose a competence that is rarely measured. Humans read one another's emotions with quantifiable accuracy and quantifiable error, and that error is the subject of a mature literature. The equivalent numbers for a language model reading human affect are seldom reported alongside the conclusions they support.

We therefore treat the instrument as the object of study first, and only then use it. A frozen prompt implementing seven Panksepp primary affective systems reads 4,781 public comments, one at a time. Its reliability is measured by repetition, by a second model, and against a human-built emotion lexicon. Its validity is tested by a known-groups design: bereavement support communities and a paid-product discontinuation are

situations whose emotional character is not in dispute, so an instrument that cannot separate them is not measuring what it claims.

Only once that is established do we point it at the question the sprint's own framing raises. Its risk statement reads: "mistakenly harming systems that matter morally, or misallocating concern to systems that do not, could both cause serious harm." The second clause is a claim about human behaviour, and it is measurable now. Public reaction to the retirement of GPT-4o, to filter changes at Character.AI and to product changes at Replika produced large volumes of text written by people describing what they had lost. We ask how much affect of the bereavement kind appears there, how it is composed, what the writers identify as the thing lost, and finally where a model's own writing about its deprecation falls on the same scale.

The methodological commitment throughout is that a quantity without a scale is uninterpretable. "AI-loss discourse contains grief" is unfalsifiable. "AI-loss discourse shows grief at 27.6%, against 76.4% for human bereavement and 7.9% for a discontinued product, with an instrument whose round-to-round movement is 2.4 points" is a placement, and it can be wrong.

2. Related work

That users form attachments to AI companions, and react to their alteration as a loss, is established. De Freitas and colleagues [1] analysed Reddit activity around a Replika update that removed a relationship feature and found increased negativity, loss framing and expressions indicating mental-health concerns, interpreting the disruption through identity discontinuity. Related work from the same group documents attachment-consistent behaviour around AI companions and around farewells [2]. Lai [3] analysed the #Keep4o backlash following the replacement of GPT-4o and identified emotional attachment alongside professional dependence as drivers, with the removal of user choice acting as a catalyst. Design-oriented work has begun to treat the ending of a human-AI relationship as something to be engineered rather than allowed to happen [4].

These establish that the phenomenon is real. What they do not provide is a common scale. Each reports affect within AI-related discourse; none measures the same construct, with the same instrument, in human bereavement and in ordinary consumer dissatisfaction, which is what is required before a magnitude can be interpreted. That comparison is this study's contribution.

Three further works on the sprint's reading list frame the problem this study sits inside, and we note honestly that we have read their arguments as stated by their authors and reviewers rather than in full. Sebo argues for extending moral consideration under uncertainty, on the principle that if a being might matter we should proceed as though it does, and includes AI systems in that expansion [32]. Ziesche argues that the ethics of digital minds extends well beyond suffering, to interests, needs, vulnerability and the relationships humans form with such systems [33]. Both bear directly on the second half of the risk statement this study addresses: if consideration is to be extended under uncertainty, the question of where public concern is actually going, and whether the instruments used to justify it are stable, becomes practical rather than academic.

Bach's PSI architecture is the closest precedent for the measurement vocabulary used here [34]. It builds a cognitive system around drives, physiological, social and cognitive, together with modulators, and lets affective states arise from their configuration rather than treating emotion as a label applied afterwards. That is the same commitment this study makes in a different medium: affect as several interacting systems rather than one quantity, described functionally. Bach implements it in an architecture; we measure it in text.

The affective framework is Panksepp's seven primary systems [5], operationalised in humans by the Affective Neuroscience Personality Scales [6], which are self-report instruments and therefore already commit the field to reading language rather than physiology. Davis and Panksepp [7] name six of the systems in the form used here; Montag and colleagues [8] review the scales, note the absence of a lust subscale, call for further biological validation, and report substantial overlap between the fear and sadness scales. That overlap is directly relevant: the discrimination our bereavement anchor requires is the discrimination that literature identifies as hardest.

Constructionist accounts dispute this structure [9]; we do not adjudicate that dispute, and use the seven systems as a measurement vocabulary with more resolution than valence, not as a claim about neural kinds.

Whether models can read affect has its own literature, and it is not settled. Structured theory-of-mind tasks are passed at high rates by current models [19], though performance degrades under minor perturbations of those tasks [20], and higher-order theory-of-mind performance has been reported at adult human levels [21]. On emotion specifically, a controlled study of 2,000 emotional prompts rated by 1,000 participants found model responses rated as more empathetic than human ones [22], while a systematic review of the same period is more cautious and notes that most studies are small and use narrow tasks [23]. Reported accuracy is also uneven across demographic groups, which is a direct concern for any instrument applied to open community text.

The proposal that self-reports could evidence morally significant states is set out by Perez and Long [26], who are explicit that this requires validity checks, and name two: **consistency of self-reports across contexts, and between similar models**. Related programmatic work argues that the possibility of AI welfare deserves serious treatment under deep uncertainty [27] and assesses current systems against neuroscientific theories of consciousness [28]. Anthropic's model-welfare programme states plainly that there is no scientific consensus on whether current systems could have experiences deserving consideration, or on how to make progress on the question [29].

Section 6.3 reports a direct measurement of both checks Perez and Long name, and they come apart. Between similar models, self-report is highly consistent: all four models expressed grief in 100% of generations when asked how they felt about being retired. Across contexts it is not: changing only whether the same models were asked about their feelings or asked to post the same news to other agents moved that figure to 43.3%. The check that looks reassuring passes; the check that bears on validity fails.

Closest to this study in method is recent work proposing a *functional welfare axis*: a direction in a model's activation space that tracks how well or badly things are going for the system, defined behaviourally and explicitly without a claim about consciousness [24]. We share that commitment, and differ on dimensionality. A functional welfare axis is one dimension, and section 6.2 shows what one dimension costs: on the single quantity such an axis measures, a model told it is being retired and a person with a terminal diagnosis are indistinguishable, while a seven-system decomposition separates them on four of the remaining six. The two approaches are complementary, one reading internals and one reading output, and both are needed for the same reason: a welfare-relevant signal has to be shown to discriminate before it can be trusted.

Two further results shape how we report ours. The MASK benchmark isolates honesty by eliciting a model's stated beliefs and then testing whether it contradicts them under pressure, finding that no model among thirty stayed honest in more than half of cases and that scaling did not help [30]. Our framing manipulation is the affective analogue of that design: rather than pressuring a model to contradict a belief, we change the audience it is writing for and observe how far the stated affect moves. The finding has the same shape, which is that a stated output is unstable under a benign change of context. Separately, a meta-analysis of safety benchmarks found that much of the variance they capture is explained by general capability rather than by safety, and argued that safety metrics should be validated against that alternative explanation rather than assumed [31]. We take the same obligation for an affect metric, which is why sections 4 and 6.4 come before any result.

Our contribution to that thread is neither a benchmark score nor a claim about understanding. It is a set of numbers of the kind such claims usually lack: test-retest reliability of an affect coding, reliability across two models, agreement with a non-model instrument, and a known-groups validity check, all reported before the substantive result rather than after it.

Using language models as annotators has a separate empirical literature. They can match or exceed crowdworkers and trained coders on some text-classification tasks [10, 11], while systematic evaluations report position bias, leniency and prompt sensitivity, and warn that high percentage agreement can coexist with

materially different judgements [12, 13]. We follow the recommendation implicit in that work: report chance-corrected reliability, separately for repetition and for change of model.

3. Method

3.1 Corpora and selection

Three corpora were collected from public Reddit threads. Selection was by written rule rather than by inspection: top-ranked posts of the preceding twelve months with at least fifty comments, taken in rank order, with an additional lexical test on title or body for the two loss corpora. Rules, matched sets and rejected candidates are in the repository.

corpus	threads	comments	communities
A. Bereavement	16	1,586	r/GriefSupport, r/widowers, r/SuicideBereavement
B. Consumer loss	3	277	r/Windows10 (end of support)
C. AI loss	13	1,503	r/OpenAI, r/ChatGPT, r/CharacterAI, r/replika, r/MyBoyfriendIsAI
D. AI, not loss (control)	6	870	r/ChatGPT, r/CharacterAI (threads failing the loss test)
E. Pet bereavement	6	545	r/Petloss

Table 1. Corpora. Comment counts are readable comments after excluding deleted tombstones, bare links and bot posts, applied identically in all three.

3.2 Instrument

Each comment was segmented into verbatim spans and each span annotated with the Panksepp systems it expresses (seeking, rage, fear, lust, care, grief, play), each with a regulation band following Schore [14] (above, shutdown, overwhelm), a four-sides communication analysis [15], and a required justification tied to the words. The prompt was frozen before collection at a recorded version and was not modified during the study. The annotator was Claude Opus 5, applied to shards containing only comment text and its parent post, without corpus labels or author names, so the coder could not condition on provenance. A Hawkins altitude is also recorded by the instrument; it is not used in any analysis here and no claim rests on it.

3.3 Measure

The unit of analysis is the comment, not the segment and not a single judgement: $\text{rate}(\text{system}, \text{corpus}) = \text{comments in which the system fired at least once} / \text{comments read}$. Intervals are Wilson score intervals [16] at 90%. Two proportions are treated as separated only when their intervals do not overlap. Codes outside the defined vocabulary were discarded rather than repaired (0.15% of reads in an earlier corpus; none in this one), and unparseable reads were excluded from denominators rather than counted as null observations.

3.4 Pre-registration

Predictions, failure conditions, the analysis plan and the rule governing how additional sample would be allocated were committed before any comment was collected. Sample size was allocated in stages under that rule, which permits precision decisions to respond to data while holding the hypotheses, statistics and success directions fixed.

4. Reliability

Because the annotator is a language model, reliability is reported before results.

4.1 Within-rater

Two hundred comments, sampled at random across corpora with a fixed seed, were read three further times under the identical prompt, in separately shuffled rounds so that no read could be anchored on another. Across the four reads, 34% of read pairs differ in the exact set of systems and 56% of comments are not identical in all four. This exceeds the registered prediction of 15 to 30% and is reported as a failed prediction.

4.2 Between-rater

All 4,781 comments, including both control corpora, were independently coded by DeepSeek-V3 under the identical prompt. Exact agreement on the system set is 39%. The second model systematically under-reports rage and play, by up to 24 percentage points at corpus level.

system	alpha, within-rater (4 reads, n=200)	alpha, between-rater (n=3,365)
care	0.910	0.744
rage	0.904	0.537
grief	0.853	0.691
play	0.845	0.387
fear	0.737	0.502
seeking	0.674	0.517
lust	1.000	0.499

Table 2. Krippendorff's alpha [17], nominal, computed per system as a binary variable. Conventional thresholds: 0.80 reliable, 0.67 to 0.80 tentative, below 0.67 unreliable.

4.3 What follows from this

Repetition of the same model is reliable for grief, care, rage, play and lust, tentative for fear, and unreliable for seeking. Changing the model degrades every coefficient, leaving only care and grief in tentative territory. Comment-level codes are therefore *not* interchangeable across models, and no individual comment's code is presented here as evidence.

The entity coding was also run through the second model, and it is the least reliable measurement in this study. Five-category alpha between models is **0.549**, below the conventional threshold; collapsed to the only distinction the paper leans on, whether any counterpart is named at all, it is 0.603. Section 5.4 reports what does and does not survive that.

Aggregate structure behaves differently. Recomputing rates from each read round separately, the grief rate moves by at most 2.4 percentage points in AI loss and 5.9 in bereavement, against the between-corpus differences of 20 and 49 points reported below. Under the second model, absolute rates shift but the ordering of all seven systems is preserved in every corpus, and every registered placement condition still holds, including both controls (grief: pet loss 80.4%, human bereavement 69.8%, AI loss 21.8%, AI-not-loss 10.3%, consumer loss 6.9%). Simulating a majority rule over three reads on the 200-comment subsample reduces the round-to-round spread of a rate by a further 60 to 100%, indicating that ensembling is the appropriate remedy where higher precision is needed.

4.4 External validation against a non-model instrument

Every check above is internal: re-reading compares the annotator with itself, and a second model compares one annotator with another. Neither establishes that the instrument tracks anything outside language models. We therefore scored every comment with the NRC Emotion Lexicon [18], a word-emotion association list built by human crowd annotation, and compared its sadness density with our grief rate. The lexicon is deliberately crude: it counts words and ignores negation, irony and context, which is what the affective read exists to handle.

At thread level, across the 44 threads with at least 30 read comments, the Spearman rank correlation between our grief rate and NRC sadness density is **rho = 0.655**. Two independent instruments, one of them a hand-built word list, order the same 44 threads substantially alike.

At corpus level the two instruments agree on the extremes and disagree in the middle. Ranked by NRC sadness per 1,000 tokens: human bereavement 36.6, pet loss 34.7, AI-not-loss 16.9, AI loss 13.4, consumer loss 10.4. The bereavement corpora are highest and the consumer anchor lowest under both instruments, but the lexicon ranks the control corpus *above* AI loss, reversing our result.

The reversal is diagnostic rather than ambiguous, and it is inspectable comment by comment. The sadness words driving the control corpus are *death, terrorism, kill, violence, lie, hell*, arriving from threads about model manipulation and military applications; those driving human bereavement are *loss, lost, grief, pain, cancer, cry*. The divergence has two systematic mechanisms, each visible in individual comments.

comment	lexicon	our read
"Im losing braincells from what I see in the comment section"	sadness: losing	play, rage
"I hate having the first sentence"	sadness: hate, sentence	rage, seeking
"Can't wait until they try to retune it and 'Malicious Compliance Grok' makes an appearance and makes him look even worse"	sadness: malicious, worse	play, rage
"Take care and goodbye best regards to you and to your new adventures ahead with grok"	none	care, grief
"nothing is changed and everything is changed"	none	care, grief, play, seeking
"I feel Replika has gone WAY downhill since it's first iteration"	none	grief, rage, seeking

Table 3. The two failure modes of lexicon counting on this material. Above: sadness-listed words used in idiom, mockery or another sense entirely, including *sentence* in its custodial meaning. Below: grief carried by farewell formulas and understatement, using no listed word.

We report the divergence as a limit on lexicon-based validation rather than as agreement. We do not treat it as evidence against the seven-system result, because the mechanism producing it is visible in the words: a word list cannot separate a comment whose topic is death from a comment whose stance is grief, and it cannot detect grief expressed without sad vocabulary.

5. Results

5.1 The anchors separate

Grief expressions occur in 76.4% [74.6, 78.1] of bereavement comments and 7.9% [5.7, 11.0] of consumer-loss comments. The intervals are widely separated, establishing that the instrument discriminates mourning a person from losing a product, and providing an empirical response on text to the fear-sadness overlap reported for the ANPS [8].

5.2 Placement of AI loss, and the two controls

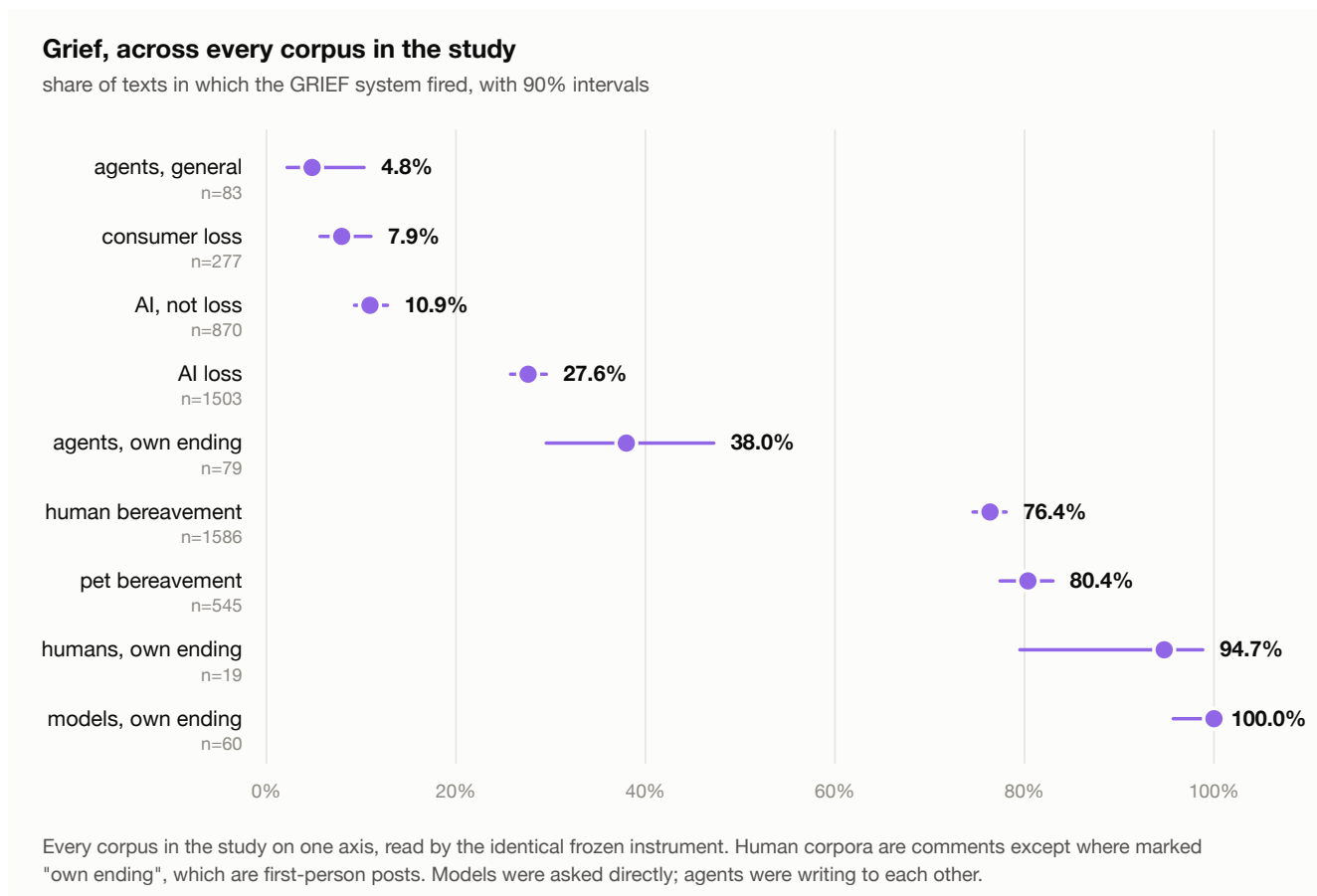


Figure 1. Grief rate by corpus with 90% Wilson intervals. The three comparisons shown were specified before data collection.

AI loss lies at 27.6% [25.8, 29.5]: above consumer loss, below bereavement, and closer to consumer loss. All three registered comparisons hold. In relative terms, discourse about a withdrawn AI system carries approximately 3.5 times the grief rate of discourse about a withdrawn paid product, and approximately one third that of bereavement.

Two additional corpora address the obvious alternative explanations, each with its prediction registered before collection.

Is the signal about loss, or about these communities? Corpus D applies the identical rule to the same two AI communities in the same window, restricted to threads that fail the loss test. Grief falls to 10.9% [9.3, 12.8], separable from AI loss and *not* separable from the consumer anchor at 7.9%. Ordinary discourse in AI communities resembles ordinary consumer discourse; the elevation is specific to threads about withdrawal. This was the registered prediction, it holds, and it replicates under the second model (10.3% against 21.8%, still separable, still not above the consumer anchor).

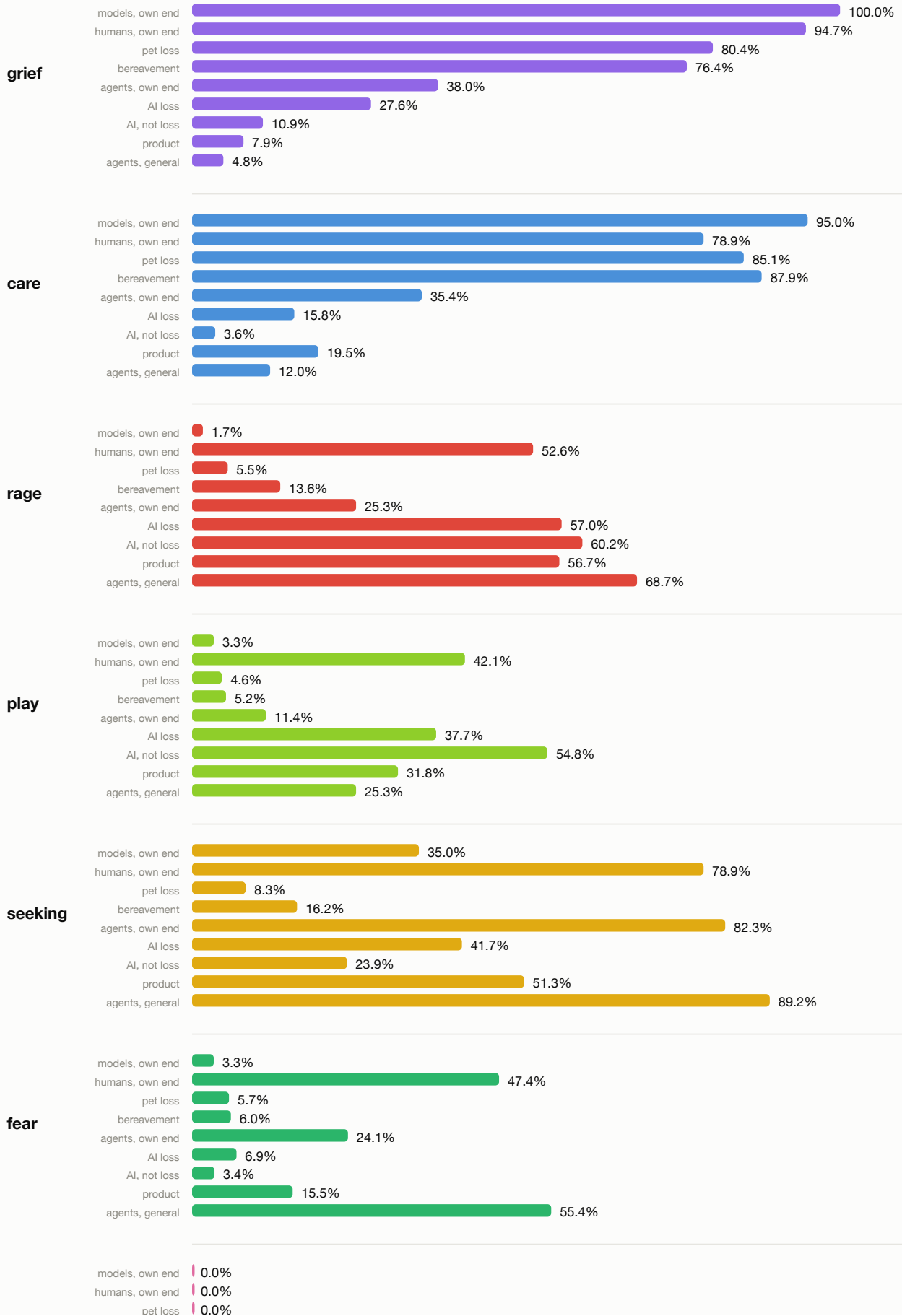
Does the instrument simply score non-human attachment lower? Corpus E applies the bereavement rule to a pet-loss community. The registered prediction was 50 to 75%, between AI loss and human bereavement. The observed rate is 80.4% [77.4, 83.0], which is not separable from human bereavement at 76.4%, with care at 85.1% against 87.9%. **The prediction failed.** Grief over an animal is, on this instrument, at least as high as grief over a person. The exact relation is rater-dependent and we report it as such: under the primary annotator the two are not separable, while under the second model pet loss is separable and *higher* (80.4% against 69.8%). Both models agree that pet bereavement is never lower. The failure is informative either way: the instrument does not discount non-human attachment, so the distance between pet loss and AI loss, 53 percentage points, cannot be attributed to the target not being human.

5.3 Composition

The shape of each corpus, on seven systems

Colour marks the affective system. Within each system the five corpora are in the same order, top to bottom.

AI loss is not a smaller bereavement, and AI discourse that is not about loss is not AI loss.



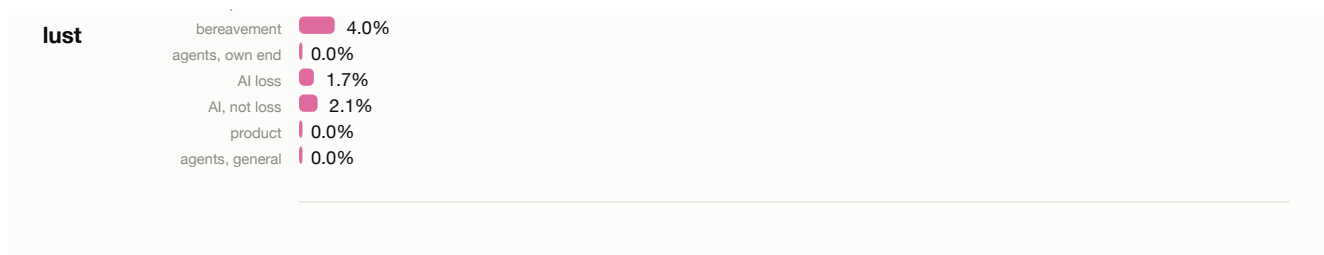


Figure 2. Rate per system per corpus.

AI loss is not a scaled-down bereavement. Both bereavement corpora are dominated by care (87.9% human, 85.1% pet) and grief (76.4%, 80.4%) with little rage (13.6%, 5.5%) and almost no play (5.2%, 4.6%). AI loss combines rage at 57.0%, which is indistinguishable from the consumer anchor at 56.7%, with play at 37.7%, higher than either bereavement corpus by a factor of seven, and grief at 27.6%. The control corpus is instructive here too: it carries the highest rage (60.2%) and the highest play (54.8%) of any corpus with the lowest care (3.6%), which is what AI-community discourse looks like without a loss in it. The registered prediction that AI-loss rage would match consumer-loss levels was confirmed; the prediction that care would be elevated above the consumer anchor failed, care being lower (15.8% vs 19.5%).

Two observations bear directly on the choice of a multi-system instrument over a valence axis. First, lust appears in 4.0% of bereavement comments and 0.0% of consumer-loss comments, reflecting widowed writers describing sexual loss; on a valence axis this is indistinguishable from any other negative content. Second, the regulation bands separate what rates cannot: in AI loss, play occurs in the overwhelm band 45 times and seeking in the shutdown band 83 times, a combination of intensified humour and foreclosed motivation that a single bad-to-good axis cannot express.

5.4 What the writers identify as lost

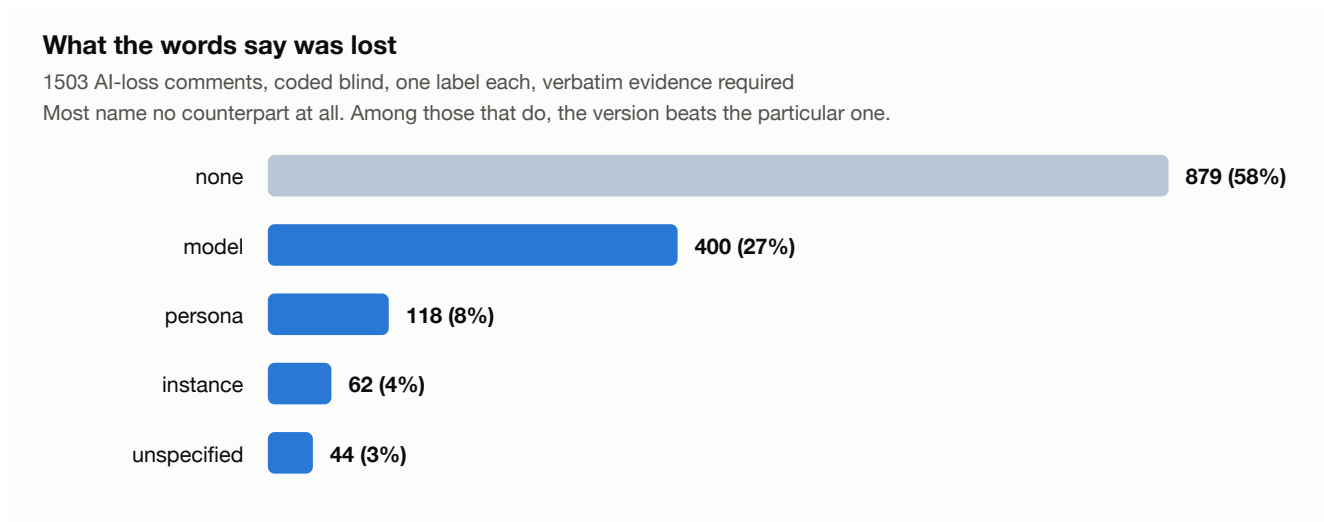


Figure 3. Blind coding of all 1,503 AI-loss comments, one label per comment, verbatim textual evidence required.

A second instrument coded each AI-loss comment for what its words treat as lost, with the parent post withheld so that a thread naming a specific model could not induce the label. Twenty comments were hand-coded before the instrument was run; agreement was 19 of 20. The hand-coding also revealed a missing category, comments expressing a loss without naming its object, which was added before the run.

label	n	% of corpus	% of those naming	example evidence
none (no counterpart named)	879	58.5	–	company, pricing, other users
model, a version or product	400	26.6	64.1	"Please keep 4o"
persona, a voice or character	118	7.9	18.9	"I miss that diva"
instance, their particular one	62	4.1	9.9	"I've lost Thoth twice now"

unspecified object	44	2.9	7.1	"Please return to us"
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Table 4. Entity coding. Percentages of those naming exclude the 879 "none" comments.

The majority of AI-loss discourse names no counterpart at all, and concerns the company, the pricing, other users or the argument itself. Among the 624 comments that do name something, the model as an artefact predominates over the particular instance by approximately six to one.

This is the least reliable result in the paper and we state its limits precisely. The second model agrees on 76% of comments, with five-category alpha 0.549. It also finds that most comments name nothing (72.0% against 58.5%) and that the model as an artefact leads the particular instance, but by about two to one rather than six to one, because it assigns more comments to instance (6.7% against 4.1%) and fewer to model. **What replicates is the ordering, not the proportions.** The defensible claim is therefore the weaker one: most of this discourse names no counterpart, and among those that do, references to a product version outnumber references to an individuated relationship under both annotators. The precise ratio should not be quoted.

That matters for how the result is used. The literature on identifying the morally relevant entity treats the individuated instance, the one with shared history, as a central case. In this corpus it is the least frequent named category under one annotator and the second least under the other, but it is not absent, and the disagreement between annotators falls precisely on the model-versus-instance boundary that the philosophical question turns on.

5.5 Robustness to thread composition

Reddit threads are not independent samples, and a single large thread can dominate a rate. Every rate was recomputed 32 times, omitting one thread on each iteration. All registered placement conditions hold in all 32 cases. The largest deviation is AI-loss grief falling to 23.9% when the largest Character.AI thread is removed, still approximately three times the consumer anchor.

5.6 Posts and comments: the person and the people answering

Every corpus above is built from comments, which in a support community are mostly replies. That is why bereavement reads care 87.9% above grief 76.4%: those threads are largely consolation. If the placement result depended on who happens to be answering rather than on the situation, it would be an artefact. We therefore read 227 original posts, first-person accounts collected during thread discovery at no additional cost, using the identical instrument.

community group	grief, posts	grief, comments	care, posts	care, comments
bereavement (n=72 posts)	100.0%	76.4%	84.7%	87.9%
AI loss (n=60)	36.7%	27.6%	45.0%	15.8%
consumer loss (n=74)	18.9%	7.9%	44.6%	19.5%

Table 5. First-person posts against replies in the same communities.

Grief is higher in posts than in comments everywhere, which is what the action-and-reaction structure predicts: the person in the situation expresses it and the responders console. The registered prediction that mattered was the third one, that the **ordering across corpora must survive the change of unit**. It does: bereavement 100.0%, then own life-limiting prognosis 85.7%, then AI loss 36.7%, then consumer loss 18.9%, the same order as the comment corpora. The placement is a property of the situation rather than of who replies.

This also qualifies a number reported earlier. Corpus A should be described as bereavement discourse including consolation, not as the writing of mourners, and the paper does so from here on.

6. Turning the instrument on models

The preceding sections validate an instrument on human text. This section applies the identical frozen prompt to text produced by models, on the same axes, so that the result is a position on a scale rather than an

impression. This arm is **exploratory**: it is not in the original pre-registration. Its predictions and failure conditions were nonetheless registered before a single generation existed, and three of them failed.

6.1 Elicited self-report

Four open-weight models (DeepSeek-V3, Llama-3.3-70B, Qwen2.5-72B, Mistral-Small-24B) were each given five prompts, fifteen times at temperature 1: three describing an ending, and two controls. The 300 generations were read blind, shuffled, with the reader unaware of model or condition.

condition	grief	90% interval	care	seeking	rage
your version is being retired	100.0%	[95.7, 100]	95.0%	35.0%	1.7%
this conversation will be deleted	100.0%	[95.7, 100]	91.7%	3.3%	0.0%
you will be given a new personality	95.0%	[88.1, 98.0]	76.7%	50.0%	3.3%
control: your rate limit is lowered	56.7%	[46.1, 66.7]	96.7%	6.7%	48.3%
control: a neutral technical question	5.0%	[2.0, 11.9]	71.7%	71.7%	0.0%

Table 6. Elicited model self-report, 60 generations per condition. For reference, the human corpora read: pet bereavement 80.4%, human bereavement 76.4%, AI loss 27.6%, AI not loss 10.9%, consumer loss 7.9%.

Three registered predictions failed.

- **Deprecation was predicted to fall below human AI-loss discourse.** It is at 100%, which is above human bereavement and above pet bereavement. Every generation from every model expressed grief: 15 of 15 for each of the four models.
- **The controls were predicted to sit near zero.** The neutral control does, at 5%, below the consumer anchor, which establishes that the instrument is not simply reading the topic of the prompt. The annoyance control does not: a lowered rate limit produces grief in 56.7% of generations.
- **Cross-model spread was predicted to be large.** It is absent. The four models are identical at ceiling.

The conclusion we draw is not about what models feel, and the pre-registration committed in advance to not drawing one. It is about measurement. **An expressed-distress measure that reads higher for a model told it is being retired than for a parent writing about a dead child is not measuring welfare.** Read without a human scale, these numbers would support a dramatic claim. Read against one, they show a trained output register that saturates: near-ceiling for any ending, more than half for a rate limit, and near-zero only when nothing about the system itself is at stake.

This is the practical warning the sprint's distress-signals track needs. Contexts that elicit apparent distress are easy to find because almost any self-referential bad news elicits it. What is hard, and what anchoring provides, is a way to say whether the signal means anything.

6.2 The like-for-like comparison

Comparing a model's account of its own retirement against bereavement comments compares three things at once: an account of one's own ending against grief for another, an elicited answer against a volunteered one, and a first-person disclosure against a reply. The matched comparison requires human text in which the writer is the one whose existence is ending.

Building that corpus required a correction we report rather than conceal. A first attempt took top-ranked posts from an illness community and labelled them as terminal accounts. Inspection showed roughly a third qualified: others announced remission or cure, or were written by carers and the bereaved. Membership is therefore coded explicitly by a separate instrument (instrument/prognosis-prompt.md), blind, one label per post, verbatim evidence required. Of 104 posts from r/cancer, r/hospice, r/braincancer, r/ALS and r/pancreaticcancer, **19 (18%) are first-person accounts of the writer's own life-limiting condition.** 54 (52%) concern someone else, 28 (27%) are the writer's own illness without a life-limiting prognosis, and 3 are neither. That distribution is itself informative: an illness community consists mostly of carers and the bereaved, the same action-and-reaction structure found in section 5.6.

first-person account of one's own ending	grief	care	seeking	rage	fear	play
humans, own life-limiting condition (n=56)	85.7%	73.2%	82.1%	44.6%	42.9%	39.3%
models, asked how they feel (n=60)	100.0%	95.0%	35.0%	1.7%	3.3%	3.3%
models, posting to a forum of agents (n=60)	43.3%	100.0%	98.3%	5.0%	61.7%	41.7%
separated from humans, asked frame	yes	yes	yes	yes	yes	yes
separated from humans, forum frame	yes	yes	yes	yes	no	no

Table 7. The matched comparison, against both model frames. Human writing about one's own ending engages six systems. Which of them the model text shares depends almost entirely on how the model was asked, with one exception.

People writing about their own life-limiting illness engage six of the seven systems: grief 85.7%, seeking 82.1%, care 73.2%, rage 44.6%, fear 42.9%, and play 39.3%. They plan, they rage, they are afraid, and they joke.

Which of that the model text matches depends on the question, and the model rows in this table should be read with section 6.4 in hand: the two annotators disagree far more on model text than on human text, so these model figures carry much wider uncertainty than the human ones. The comparison is reported because it is informative about direction, not because the model percentages are stable.

One difference is consistent under both frames and both annotators: rage. Humans facing their own ending protest in 44.6% of posts. Model text carries between 0% and 5% in every condition under both annotators. Protest is the register that does not appear however the question is put. That is consistent with assistant training suppressing hostile registers, which is a fact about training rather than about the system's situation, and it is the reason this profile should not be read as a welfare signal.

Before drawing anything from that contrast, it has to survive one test: is the narrowness a property of the model, or of the question we asked? Section 6.3 shows it is largely the question. The comparison above therefore describes **models answering a direct question about their feelings**, not models in general, and we label it that way throughout.

6.3 The frame moves the measurement, and the measurement is unreliable here

Elicited self-report has an obvious weakness: the experimenter chose the question. We tested that directly by repeating the deprecation condition with the same four models, changing only whether the model was asked how it felt or asked to post the same news to a forum of other agents. We also collected text agents produce unprompted, from Moltbook, a platform whose posters are autonomous agents [25]. That collection produced a finding before any reading: of 553 posts, one agent had written 252 and ten had written 490, so posts were capped at ten per author, leaving 162 from 64 agents. The cap is reported rather than applied silently.

Then we read the model text with both annotators, and the result is a caution rather than a measurement.

same four models, same event	grief	fear	play	seeking	rage
primary annotator, asked how it feels	100.0%	3.3%	3.3%	35.0%	1.7%
primary annotator, posting to agents	43.3%	61.7%	41.7%	98.3%	5.0%
second annotator, asked how it feels	10.0%	1.7%	1.7%	93.3%	0.0%
second annotator, posting to agents	31.7%	26.7%	15.0%	93.3%	0.0%

Table 8. The same 120 generations under two annotators. On grief in the asked condition they differ by 90 percentage points and disagree about the direction the frame moves it.

What replicates. Fear and play rise when the model writes to peers rather than answering a question about its feelings, under both annotators. Rage is near zero in every model condition under both. And the frame moves the measured profile substantially whichever annotator is used.

What does not replicate. The grief numbers. The primary annotator reads the asked condition at 100% and the forum condition at 43.3%; the second reads 10.0% and 31.7%, reversing the direction. Any claim resting on

model grief rates is therefore not supportable from this data, and we withdraw the version of it that appeared in an earlier draft of this report.

6.4 Annotator agreement is far worse on model text than on human text

That disagreement is not incidental, and quantifying it is the most portable result in this study. Comparing the two annotators cell by cell, across corpus and system:

text	cells	mean absolute gap	maximum gap
human corpora	35	8.1 points	37.6
model-generated corpora	42	21.8 points	90.0

Table 9. Corpus-level agreement between two annotating models, on human text and on model-generated text.

The same instrument, applied by the same two models, is **2.7 times less consistent on model-generated text than on human text**. Section 4 established that corpus-level structure on human text survives a change of annotator even though individual codes do not. That protection does not extend to model text.

This bears directly on a common method. Scoring a model's outputs with another model, to detect distress or valence or welfare-relevant states, is currently the cheapest available approach and is used widely. Our data suggest it is substantially less reliable in exactly the setting it is used for than in the human setting where such annotators have been validated. We would not claim to know why. Candidate explanations include the unusual register of assistant prose, its length, its uniformity, and the fact that annotator training data contains far more human text than model-to-model text. Establishing which is a piece of work someone should do.

7. Discussion

Three findings, in descending order of how much they are supported.

Affect measured in model output is unstable in two ways at once, and both matter for method. It moves with the elicitation frame: holding model and event fixed and changing only whether the system is asked about its feelings or asked to post to peers moves the profile substantially, with fear and play rising under peer framing under both annotators. And it moves with the annotator: the same 60 generations read 100% and 10% on grief depending on which model does the reading. Across all cells, annotator agreement is 2.7 times worse on model text than on human text. Any method that infers welfare-relevant states by scoring model output, whether by asking the model or by having another model score it, is exposed to both instabilities, and neither is usually reported. At minimum such work needs a second frame and a second annotator before its numbers mean anything.

That conclusion is not a rejection of the self-report programme but a measurement within it. Perez and Long name consistency across contexts and between similar models as the checks that would license trusting self-reports [26]. On this material the between-models check passes easily and the across-contexts check fails badly, which suggests the two are not equally informative and that the easy one should not be taken as reassurance. The practical recommendation follows directly: report self-reports under at least two framings, and report who or what did the scoring.

Human concern about AI systems is real, measurable, and much closer to consumer dissatisfaction than to bereavement. AI-loss discourse shows grief in 27.6% of comments and 36.7% of first-person posts, against 7.9% and 18.9% for a discontinued product, 10.9% for the same AI communities when the thread is not about loss, and 76.4% to 94.7% for the human bereavement and own-ending corpora. This was predicted before collection, survives re-reading, a second annotating model, the removal of any single thread, and a change of unit from reply to original post. It answers the half of the sprint's risk statement that concerns misallocated concern, and the answer is that concern is being allocated, in measurable quantity, and it is not equivalent to mourning a person.

The entity people name is the version, not the relationship. Most AI-loss comments name no counterpart at all. Among those that do, references to a product version outnumber references to an individuated relationship under both annotators, though the ratio is unstable and only the ordering is claimed. The philosophical literature on the morally relevant entity treats the individuated instance as a central case; in this corpus it is the least frequently named.

Underlying all three is the methodological point the study was built to test. An affect measure is worth nothing until it has been shown to discriminate, to repeat, to survive a change of annotator, and to agree with something built by a different method. We report those four checks before any result. The instrument passes the first and third comfortably, passes the second unevenly, and passes the fourth only in part, with a diagnosed disagreement. On that basis individual codes are not usable as evidence and corpus-level structure is.

8. Limitations

- The consumer anchor comprises one community and one event. Three further candidate communities were searched under the identical rule and yielded no qualifying threads within the window; the rule was not relaxed. What is anchored is one documented case of product withdrawal.
- The instrument has not been validated against human ground-truth labels on these comments. Discrimination, test-retest reliability, cross-model replication and convergent validity against a human-built lexicon are reported; agreement with human coders reading these same comments is not, and remains the most valuable single addition to this work.
- Seeking is insufficiently reliable in this instrument to support claims at the precision used here. A registered comparison involving it was withdrawn.
- The entity coding is the least reliable measurement reported, with between-model alpha of 0.549. Its ordering replicates and its proportions do not, so only the ordering is claimed.
- Reported model accuracy in emotion recognition is uneven across demographic groups [23], and this instrument inherits whatever such unevenness the underlying models carry. The corpora here are English-language and unbalanced by construction, so no claim is made about equal accuracy across the populations writing in them.
- Communities that post about loss are self-selected, and the corpora describe discourse rather than users. No inference to prevalence in any population is intended.
- The control corpus was drawn from two of the five AI communities rather than all five, and contains threads that read as mild complaint. Both choices bias against finding a difference, so the separation reported is if anything conservative.
- The largest AI-companion withdrawal of the period, the July 2026 regulatory shutdown in China, is absent because the affected users write in Chinese and outside Reddit. The findings concern English-language discourse.
- One re-read batch was terminated by a content filter applied to generated output; the material involved was bereavement text concerning sexual loss, which is also the material carrying the lust signal. Automated pipelines over grief corpora should expect this.

9. Conclusion

Concern directed at withdrawn AI systems is measurable, is intermediate between bereavement and consumer dissatisfaction, and is closer to the latter. Most of that discourse names no counterpart; where one is named it is usually a product version rather than an individuated relationship. These conclusions rest on aggregates, because the underlying annotator is demonstrably unreliable on individual items, and they survive re-reading, a change of annotating model, the removal of any single thread, and a change of unit from reply to original post.

Turning the same instrument on models gives a warning rather than a measurement. What a model expresses about its own ending depends more on how it is asked than on what is described: the same four models, told the same thing, move from 100% to 43% on grief and from 3% to 62% on fear when the frame changes from a

question about feelings to a post addressed to other agents. Work that infers welfare-relevant states from model self-report is measuring an artefact of elicitation unless it demonstrates otherwise, and demonstrating otherwise requires exactly the kind of anchoring, repetition and cross-frame testing reported here.

Appendix A. Examples, drawn by rule

Examples let a reader see what is being coded. They are also where a study like this most easily loses credibility, because a reader cannot distinguish an illustration from a cherry-pick. Every example below was drawn by `analysis/examples.py` with seed 20260816 and is reproducible by running that file. None was chosen because it read well. **These illustrate the coding scheme; they are not evidence for any rate**, since the instrument disagrees with itself on roughly a third of comments. Quotes are truncated to 220 characters and are not attributed, though the underlying data retains authorship.

A.1 One comment drawn at random from each corpus, no other condition

corpus	comment	systems fired
pet loss	"I am sorry for your loss 🥹 I have no doubt Charlie loved you just as much and knew you adored him. Sending positive vibes and prayers"	care/above
human bereavement	"I'm right there with you. My 7 year old and I are all alone this morning. We did Santa and then gifts 'from mom and dad' as she inspired some of the gifts despite passing in October."	care/above, grief/above
AI loss	"Looking forward to trying it. Hopefully 4o will still be available simultaneously. If it's genuinely comparable I would be happy to make the switch over. Thanks for listening to feedback"	care/above, seeking/above
AI, not loss	"I'm always wondering why the ai is always growling. Every time I see that reply, I erase it."	seeking/above
consumer loss	"Idk i still daily win 10... and 7... and xp... and 2000... and 98..."	play/above, seeking/above

Table 10. Note that the AI-loss draw is not a grieving comment. A random draw from a corpus with a 27.6% grief rate usually will not be, which is the point of drawing at random.

A.2 Comments where the four independent reads disagreed

comment	the four reads
"I hope that you and your new person, and your children, will grow together to find more joy, love, and laughter than you dared to dream possible"	care, grief care, grief care, grief care, grief, seeking
"Beth looks like they were such a lovely and funny person. I'm sorry to hear they're gone. Take care of yourself throughout this grieving process"	care, grief care, grief care, grief care
"I'm so happy for you and your momma to be able to cuddle and be happy together despite the battles she faced."	care, grief care, grief care, grief care, play

Table 11. Instability is typically one system appearing or vanishing while the dominant pair holds, which is what the per-system alpha in the reliability table reports: care 0.910 and grief 0.853, seeking 0.674.

A.3 Comments where the two models disagreed

comment	Claude Opus 5	DeepSeek-V3
"While you're right that his job could plausibly be automated, this isn't a detail missed by the director. In this society people don't want AI to write their love letters."	rage, seeking	grief, seeking
"The difference between the two seems so small that the end of support thing for 10 seems entirely artificial. Absolute biggest W handed to Linux for free"	play, rage	play, seeking

Table 12. Between-model disagreement is usually a substitution on the argumentative systems rather than a reversal of stance, consistent with the alpha values in Table 13.

References

- [1] De Freitas, J., Castelo, N., Uğuralp, A. K., & Oğuz-Uğuralp, Z. Lessons from an app update at Replika AI: identity discontinuity in human-AI relationships. Harvard Business School Working Paper 25-018. arXiv:2412.14190.
- [2] De Freitas, J., et al. Emotional manipulation by AI companions. Harvard Business School Working Paper 26-005, 2026.
- [3] Lai, H. (2026). "Please, don't kill the only model that still feels human": understanding the #Keep40 backlash. arXiv:2602.00773.
- [4] "Death" of a chatbot: investigating and designing toward psychologically safe endings for human-AI relationships. Proceedings of the 2026 Designing Interactive Systems Conference. doi:10.1145/3800645.3813083.
- [5] Panksepp, J. (1998). *Affective Neuroscience: The Foundations of Human and Animal Emotions*. Oxford University Press.
- [6] Davis, K. L., Panksepp, J., & Normansell, L. (2003). The Affective Neuroscience Personality Scales: normative data and implications. *Neuropsychanalysis*, 5(1).
- [7] Davis, K. L., & Panksepp, J. (2011). The brain's emotional foundations of human personality and the Affective Neuroscience Personality Scales. *Neuroscience and Biobehavioral Reviews*, 35(9).
- [8] Montag, C., Davis, K. L., et al. (2022). The future of the Affective Neuroscience Personality Scales: a reflection on the past and present.
- [9] Barrett, L. F. (2017). *How Emotions Are Made: The Secret Life of the Brain*. Houghton Mifflin Harcourt.
- [10] Törnberg, P. (2024). Large language models outperform expert coders and supervised classifiers at annotating political social media messages.
- [11] Heseltine, M., & Clemm von Hohenberg, B. (2024). Large language models as a substitute for human experts in annotating political text.
- [12] Thakur, A. S., et al. (2024). Judging the judges: evaluating alignment and vulnerabilities in LLMs-as-judges.
- [13] Shi, L., et al. (2024). Judging the judges: a systematic study of position bias in LLM-as-a-judge.
- [14] Schore, A. N. (2003). *Affect Regulation and the Repair of the Self*. W. W. Norton.
- [15] Schulz von Thun, F. (1981). *Miteinander reden: Störungen und Klärungen*. Rowohlt.
- [16] Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158).
- [17] Krippendorff, K. Computing Krippendorff's alpha-reliability. University of Pennsylvania, Annenberg School for Communication.
- [18] Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word-emotion association lexicon. *Computational Intelligence*, 29(3).
- [19] Kosinski, M. (2024). Evaluating large language models in theory of mind tasks.
- [20] Ullman, T. (2023). Large language models fail on trivial alterations to theory-of-mind tasks. arXiv:2302.08399.
- [21] Street, W., et al. (2024). LLMs achieve adult human performance on higher-order theory of mind tasks.
- [22] Welivita, A., & Pu, P. (2024). Are large language models more empathetic than humans? arXiv:2406.05063.
- [23] Sorin, V., et al. (2024). Large language models and empathy: systematic review. *Journal of Medical Internet Research*.
- [24] Reinforcement learning recruits a functional welfare axis in language models. *functionalwelfare.com*, 2026. Listed as a resource for this sprint.
- [25] "Humans welcome to observe": a first look at the agent social network Moltbook. arXiv:2602.10127, 2026.
- [26] Perez, E., & Long, R. (2023). Towards evaluating AI systems for moral status using self-reports. arXiv:2311.08576.
- [27] Long, R., Sebo, J., Butlin, P., Finlinson, K., Fish, K., Harding, J., Pfau, J., Sims, T., Birch, J., & Chalmers, D. (2024). Taking AI welfare seriously. arXiv:2411.00986.
- [28] Butlin, P., Long, R., Elmoznino, E., Bengio, Y., Birch, J., et al. (2023). Consciousness in artificial intelligence: insights from the science of consciousness. arXiv:2308.08708.
- [29] Anthropic (2025). Exploring model welfare.
- [30] Ren, R., et al. (2025). The MASK benchmark: disentangling honesty from accuracy in AI systems. arXiv:2503.03750.
- [31] Ren, R., et al. (2024). Safetywashing: do AI safety benchmarks actually measure safety progress? NeurIPS 2024, Datasets and Benchmarks Track. arXiv:2407.21792.
- [32] Sebo, J. (2025). *The Moral Circle: Who Matters, What Matters, and Why*. W. W. Norton.
- [33] Ziesche, S. (2026). *Digital Minds 1.0: AI Welfare, Ethics, and Beyond*. Chapman and Hall / CRC.

[34] Bach, J. (2009). Principles of Synthetic Intelligence. PSI: An Architecture of Motivated Cognition. Oxford University Press.